

Summary

Software Engineer with hands-on experience in Python, Agentic Systems, and Generative AI workflows. Experienced in decomposing business problems into AI workflows, context engineering via RAG and MCP, and multi-model grounding. **6x Hackathon Winner** (including **Smart India Hackathon 2024**) and **IEEE Tech4Good Grant Project Tech Lead** that was awarded \$4000 grant for building an AI-powered app for farmers to increase crop yield.

Technical Skills

Programming Languages & Frameworks: Python, NumPy, Pandas, Matplotlib, Scikit-learn, Java, Spring Boot, SQL
AI & Agentic Systems: Generative AI, LLMs, Agentic Workflows, MCP, Prompt Engineering, LangChain, Transformers, Pydantic, Hugging Face, Tool Calling, RAG, Semantic Search, Hybrid Retrieval, FAISS, Qdrant, Embeddings, Grounding, Vector Databases
Backend & Cloud: FastAPI, RESTful APIs, PostgreSQL, Supabase, Docker, CI/CD, Git, GitHub, Azure Databricks
CS & Software Engineering: Data Structures & Algorithms, System Design, OOPs, DBMS, SDLC, Unit Testing, Integration Testing, Code Reviews, Debugging

Education

Acropolis Institute of Technology and Research

Indore, India

B.Tech. in Computer Science & Engineering (AI & ML) – CGPA: 7.43

October 2022 – June 2026

Experience

Associate Software Engineer

August 2026 – Present

ClearTrail Technologies Pvt. Ltd.

Indore, India

- Developed backend REST APIs in Java for application integration, implementing robust endpoints and optimizing relational SQL queries for Enterprise Platform.
- Authored clean, maintainable code adhering to established coding standards; actively debugged defects and collaborated in Agile sprint ceremonies and code reviews alongside QA and senior developers.

Software Development Engineer 1

May 2026 – August 2026

Vected Technologies Pvt. Ltd.

Indore, India

- Engineered RAG applications using LangChain, vector databases, and LLMs, performing end-to-end integration, prompt tuning, and evaluation for context-aware enterprise document search.
- Designed Voice AI agents with multi-step reasoning, API tool-calling, and context engineering; decomposed complex workflows to improve business efficiency.

Projects

Pyrrhon – Voice First Codebase Agent

pyrrhon.vercel.app

- Architected voice-first engineering agent for codebase reasoning and system design (PRDs, HLDs), orchestrating multi-LLM workflows across Groq, OpenAI, Anthropic, and Gemini with automated fallback routing.
- Engineered context grounding gates requiring deterministic file:line citations to eliminate hallucinations, paired with dynamic token-budget management and session persistence.
- Integrated Model Context Protocol (MCP) tools and low-latency voice pipeline (STT/TTS, turn detection, barge-in); validated with 1,200+ unit/integration tests.

Sentient – AI NPC Dialogue Platform

sentient-npc.vercel.app

- Built production RAG dialogue platform serving OpenAI-compatible SSE streaming endpoints for game NPCs, dynamically grounding generative dialogue in multi-format documents (PDF/TXT) via semantic search.
- Engineered hybrid retrieval architecture across FAISS and Qdrant Cloud (dense + sparse search) with strict multi-tenant isolation to guarantee partition security.
- Implemented strict security & governance with Fernet-encrypted credential vaults and zero-downtime key rotation; integrated Langfuse tracing to monitor token usage, cost, and latency.

Signify – Real-Time ISL Translation

github.com/prabhjot0109/signify_sih

- Developed award-winning app (National Winner, Smart India Hackathon 2024) translating Indian Sign Language into text/speech and vice-versa in real-time with 90% recognition accuracy.
- Architected scalable Python backend with FastAPI REST APIs and Flutter mobile frontend, adhering to clean OOP principles, robust data validation, and unit test suites.
- Built Random forest based ML Model with MediaPipe landmark tracking algorithms, for hand gesture recognition, achieving 200ms latency for real-time bidirectional communication.

Honors & Awards

- IEEE Tech4Good Grant Project Tech Lead:** Served as tech lead for the \$4,000 IEEE HTB grant-funded project focused on humanitarian tech innovation.
- 6x Hackathon Winner:** National Winner – Smart India Hackathon (SIH) 2024, 1st Place – Intellify 3.0, 1st Place – HackWave, 2nd Place – Code for Bharat 2, 2nd Place – Codespire, 4th Place – Prayatna.